MCO-1D Test Cards

TC-01 Bench · TC-02 Captive Hover · TC-03 Truck Fixed-Wing · TC-04 First Free Flight

Complete procedures for all four commissioning test cards.

Document	MCO-1D_Test_Cards_v24
Revision	September 2026
New in v24	Full content TC-01 through TC-04 (stubs replaced with complete procedures)
Audience	Commissioning engineer — engineering background required

TC-01 Bench Check

Objective	Verify all systems respond correctly on the bench before any hover attempt. Confirm wing guard geometry, ODri
Prerequisites	Aircraft assembled. Battery charged >22V. GS laptop on same network. Fuselage width and TPU start measure
Materials	Aircraft (no propulsion hazard clearance needed — wings not spinning fast). Laptop with GS. Caliper (for wing g

■ **CAUTION** Do not attempt TC-02 with any faulted axis or unverified wing guard geometry.

Procedure

Step 1. Connect and verify GS

Start GS launcher. Confirm green CONNECTED badge. Run: status — verify all axes report ready, fault_count = 0.

Step 2. Enter fuselage geometry

wc_set_fuselage_width [measured mm]. wc_set_tpu_start [measured mm]. wc_geometry_report — verify TPU zone tip separation > 0mm at 96° amplitude.

Step 3. ODrive calibration check

GS: soft_stop_status — verify all 8 axes homed. If any axis shows fault, run per-axis calibration sequence. Do not proceed with any faulted axis.

Step 4. Sensor verification

GS: pitot_status — verify airspeed reads ~0 m/s (bench, no airflow). GS: attitude_hold_status — verify roll and pitch < 2° offset. GS: battery_status — verify voltage matches multimeter reading.

Step 5. Wing guard functional check

Command low-amplitude sweep (20° each wing, 0.5 Hz). Verify wing guard reports all 8 wings active with no interventions. Increase to 60° amplitude. Verify guard still reports clean. Stop wings. Verify position hold restores.

Step 6. Slow stroke test

Arm aircraft (all personnel clear). Command 1.0 Hz, 50° amplitude for 10 wingbeats. Verify no mechanical binding, no rope slack, no ODrive fault. Disarm.

Step 7. Knowledge check

GS: knowledge_boot_check — verify knowledge directory accessible. GS: disc_registry — verify 97 hypotheses loaded, queue = 97. GS: mechanism_weight_report — verify 6 genome axes present.

Step 8. Export and verify

GS: tc01_confirm — marks TC-01 complete in test record. GS: tc_export — export test results to file.

Pass criteria

- ■ All 8 ODrive axes homed with no faults
- ■ Fuselage width and TPU start entered and geometry verified
- ■ Wing guard reports clean at 60° amplitude
- ■ Pitot reads ±0.2 m/s at bench
- ■ Attitude offset < 2° on both axes

- ■ Battery voltage matches multimeter within 0.1V
- ■ Knowledge directory accessible, 97 hypotheses queued
- ■ Slow stroke test: no fault, no binding, no slack

Tested by	_____
Date	_____
Result	PASS / FAIL / CONDITIONAL
Notes	

TC-02 Captive Hover

Objective	Verify stable hover with all 8 wings at minimum safe altitude. Characterise hover trim, validate wing guard in fligh
Prerequisites	TC-01 passed. Clear area 3m radius, 2m height. Tether rated >50N attached to aircraft hard point and ground a
Materials	Aircraft with tether. GS laptop. Tether: 1.5m Dyneema or steel, rated >50N, with quick-release. Tape measure. V

■ **CAUTION** Tether must be taut before arming. Observer must have clear abort signal. Stop immediately on any structural sound or ODrive fault.

Procedure

Step 1. Safety setup

Attach tether to aircraft hard point. Verify quick-release accessible. Clear 3m radius. Brief observer: observer monitors attitude and calls "abort" if roll >20° or pitch >30° or any structural sound.

Step 2. Arm and initial spool-up

Arm aircraft. Command 0.5 Hz, 30° amplitude. Verify wings moving symmetrically. No faults. Observer checks all 8 wings visually.

Step 3. Hover attempt

Increase to 1.87 Hz, 96° amplitude. Aircraft should lift off tether. If not: check weight, increase amplitude to 108° max. Target altitude: 0.3–0.5m AGL.

Step 4. Hover stabilisation

Hold hover for 2 minutes. Monitor: VSI < ±50 ft/min hands-off, roll < 5°, heading drift < 10°. Record motor current, bus voltage, rope health at 1-minute intervals.

Step 5. Wing guard validation

GS: wc_geometry_report — verify in-flight tip separation. GS: wing_guard_status — verify guard event count per minute. Acceptable: < 1 guard event per minute at hover amplitude.

Step 6. Flight monitor baseline

GS: monitor_report — record trim baseline VSI and roll. Note rope life rate (%/min). Must be < 2%/min.

Step 7. Vario check

GS: vario_mode netto — verify netto reads ~0 in still indoor air. GS: soaring_status — verify polar sink reads ~0.53 m/s estimate. Return to vario_mode basic.

Step 8. Discovery check

GS: observer_status — verify observer active. Enable discovery for 2 minutes. GS: disc_registry — verify first evaluation has completed. GS: mechanism_weight_report — verify genome shows any change from equal.

Step 9. Land and disarm

Reduce amplitude gradually. Land softly. Disarm. GS: tc02_confirm — marks TC-02 complete. GS: persist_checkpoint — save all learning.

Pass criteria

■ Aircraft achieves stable hover at 0.3–0.5m AGL

- ■ Hover maintained 2 minutes: VSI < ±50 fpm, roll < 5°
- ■ Wing guard < 1 event/minute at hover amplitude
- ■ Rope life rate < 2%/min
- ■ Netto variometer reads ~0 in still air
- ■ First discovery evaluation completed
- ■ No ODrive faults during hover
- ■ Learning saved successfully

◆ **NOTE** Hover trim will improve over multiple TC-02 sessions as the genome learns. Expect better hands-off stability by the 3rd session.

Tested by	_____
Date	_____
Result	PASS / FAIL / CONDITIONAL
Notes	

TC-03 Truck Fixed-Wing

Objective	Verify fixed-wing forward flight on a moving vehicle at 8-12 m/s (16-23 kt). Characterise the glide polar, validate
Prerequisites	TC-02 passed. Calm day (wind < 5 m/s). Open area, vehicle speed 8-12 m/s. Pilot familiar with hover controls. S
Materials	Aircraft. Vehicle (truck, trailer, or boat). Restraint line for low-speed tests. GS laptop (can use hotspot if no netwo

■ **CAUTION** Keep aircraft within safe glide range of vehicle. Do not exceed vehicle speed rating. Abort if aircraft rolls more than 30° without pilot input.

Procedure

Step 1. Vehicle and safety setup

Vehicle moves at 8 m/s (28 km/h) on straight, clear path. Observer monitors aircraft attitude from vehicle. Abort call = vehicle stops, pilot kills motor.

Step 2. Hover transition

Begin in hover. As vehicle accelerates to 5 m/s, aircraft transitions to forward flight automatically. Monitor VSI and heading through transition.

Step 3. Cruise establishment

At 9-10 m/s vehicle speed, establish level cruise. Record: motor current, bus voltage, heading stability, VSI variation. Note lever position for altitude hold (should be near 50%).

Step 4. Glide polar measurement

Reduce power to 5%. Allow aircraft to descend at cruise speed. GS: vario_mode netto — note netto sink rate at 9 m/s, 10 m/s, 11 m/s. These three points characterise the polar. GS: soaring_status — record measured values.

Step 5. Stability check

At cruise, release roll stick for 10 seconds. Record heading drift rate. Apply 15° bank, release stick — measure return rate. These values feed H-DIH-5 spiral stability assessment.

Step 6. High-speed test

If conditions allow: increase to 15-18 m/s. Verify no oscillation. Record power and VSI at this speed. GS: speed_performance_status — verify sprint transition.

Step 7. Land and save

Reduce speed to hover transition. Land at vehicle stop. GS: tc03_confirm. GS: persist_checkpoint.

Pass criteria

- ■ Aircraft maintains stable fixed-wing cruise at 9-10 m/s
- ■ Polar measured at 3 speeds — netto values recorded
- ■ Heading drift hands-off < 5°/5s (spiral stability baseline)
- ■ Pitot airspeed matches vehicle speed within 0.5 m/s
- ■ Lever ~50% gives near-level cruise (altitude hold baseline)
- ■ No oscillation at any tested speed
- ■ Learning saved

◆ **NOTE** The netto readings from this card update the polar estimates. Future MacCready advisories will be more accurate.

Tested by	_____
Date	_____
Result	PASS / FAIL / CONDITIONAL
Notes	

TC-04 First Free Flight

Objective	First untethered, unsupported flight. Verify all systems in real free-air conditions. Characterise actual handling qu
Prerequisites	TC-02 and TC-03 passed. Calm day (wind < 5 m/s). Open area 50m radius clear. VLOS maintained at all times.
Materials	Aircraft with charged battery. GS laptop. Video camera. Spare battery. Emergency stop within reach.

■ **CAUTION** VLOS must be maintained at all times. If aircraft becomes unresponsive to inputs: kill motor immediately — aircraft auto-returns. Do not fly near people, roads, or buildings until TC-04 passed.

Procedure

Step 1. Area and safety brief

Walk the area. Confirm no hazards. Brief observer: watch for any abnormal attitude or descent rate. Abort call: pilot kills motor, aircraft auto-returns. Establish comm protocol before arming.

Step 2. Pre-flight and arm

"Ready to fly" — confirm all checks pass. Confirm battery > 22V. "Arm it" — step back 3m from aircraft before arming. Confirm badge amber.

Step 3. Initial hover

Climb to 3m AGL. Hover for 1 minute. Note hands-off behaviour: VSI < ±50 fpm, heading drift < 5°/min. GS: monitor_report — confirm no Tier 1 or 2 alerts.

Step 4. Gentle forward flight

Transition to forward flight at 7-8 m/s. Fly a rectangle: 2 legs into wind, 2 crosswind. Note coordination in turns — any sideslip tendency? Note hands-off heading stability on straight legs.

Step 5. Discovery run

On a straight leg at cruise: enable discovery. Fly straight for 3 minutes. GS: disc_registry — note first evaluations. Observer will propose within 2 minutes if trim drift detected.

Step 6. Soaring check

Reduce power to < 15% for 30 seconds. GS: soaring_status — note netto reading. In calm air should read < ±0.1 m/s. Restore power. Note transition smoothness.

Step 7. Stability check

At 9 m/s, release roll stick for 5 seconds on a straight leg. Note heading drift. Enter 15° bank, release — note return tendency. Record both for H-DIH-5 baseline.

Step 8. ALSS Class A sortie

"Start practice flights" — aircraft begins Class A ALSS. Monitor from GS. First sortie typically 8-12 minutes. Pilot may intervene at any time.

Step 9. Land and save

"Land" or reduce power to hover and descend manually. After touchdown: "We're done". Confirm "Session saved. Safe to disconnect." before battery disconnect. GS: tc04_confirm. GS: tc_export.

Pass criteria

- ■ Aircraft achieves stable free hover at 3m AGL
- ■ Transitions smoothly to forward flight
- ■ Rectangular flight pattern completed without pilot difficulty
- ■ No Tier 1 flight monitor alerts during flight
- ■ First discovery evaluation completed during free flight
- ■ Netto variometer reads < ±0.1 m/s at power-off in calm air
- ■ ALSS Class A sortie started (and completed if battery allows)
- ■ Session saved successfully

◆ **NOTE** TC-04 pass does not mean the aircraft is fully learned — it means it is safe to fly and learning is working. Handling will improve over the first 10-20 flights as the genome converges.

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Notes	